

Wandering Black Hole Caught Feeding on Its Own Wake

A 35,000-solar-mass black hole off the galactic nucleus accretes gas captured by gravitational focusing.

arXiv:2608.10719 — Wandering 35,000-solar-mass black hole fed by gravitational wake

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Background

Intermediate-mass black holes may seed supermassive ones, yet many should wander far from galactic nuclei after hierarchical mergers. Off-nuclear black holes lack the usual fueling channels available at galactic centers.

Theory predicts wandering black holes capture gas from gravitational wakes, but direct observational evidence has remained elusive.

What this paper is selling

The authors report a 35,000-solar-mass off-nuclear black hole accreting through a gravitational wake in UGCA 320. Multi-epoch spectroscopy reveals upstream flow, a dense downstream wake, and optically thick absorbers within the capture radius.

Key results to remember

- Broad lines, compact continuum, variability, and power-law SED confirm the black-hole nature.
- Three gas components: blueshifted upstream flow, redshifted dense downstream wake, and optically thick absorbers.
- Absorbers inside the capture radius drive rapidly changing-look broad-line variability.

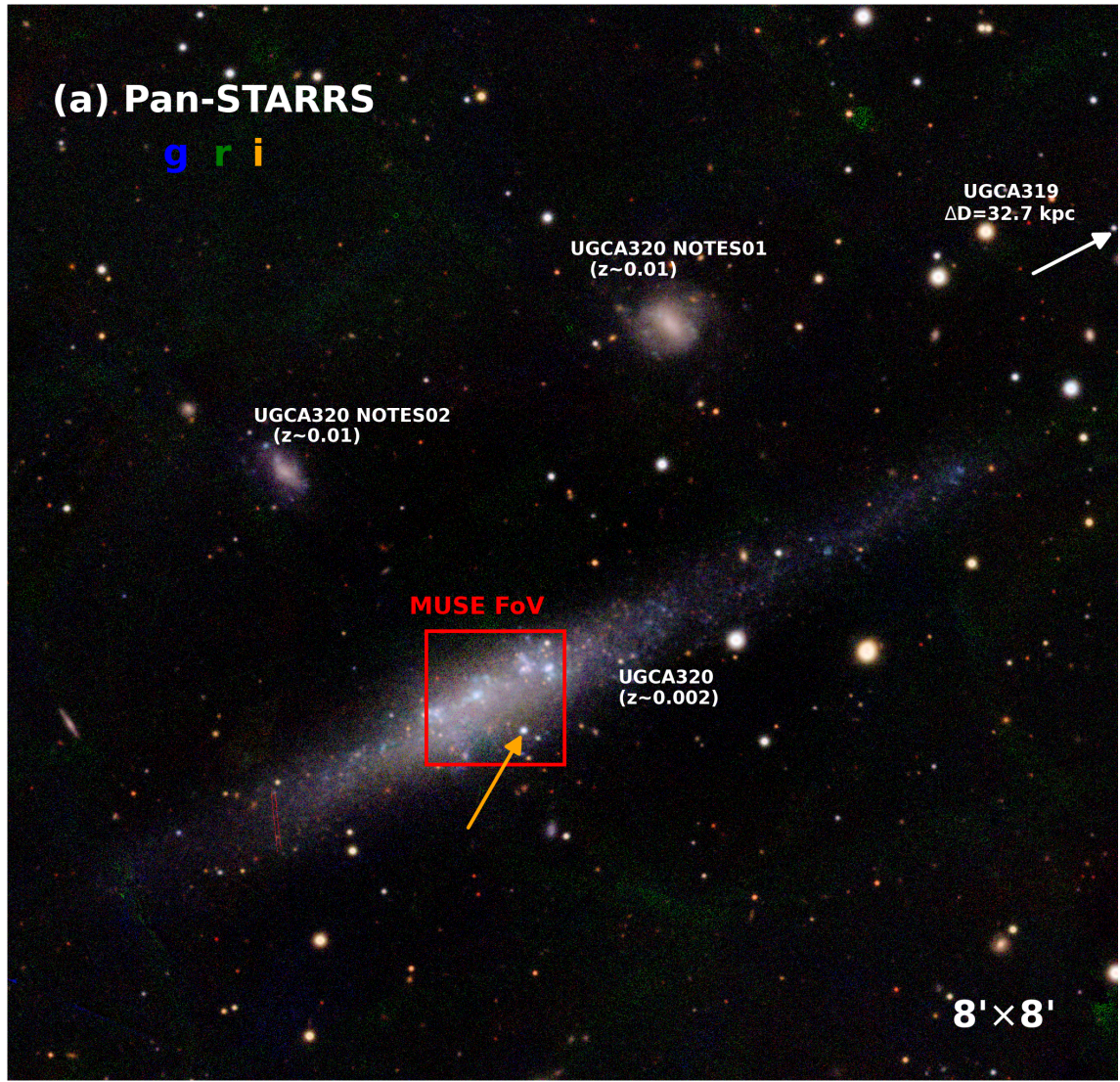


Figure 1 | HST and MUSE views pinpoint the off-nuclear black hole and map its [O III] velocity field within UGCA 320.

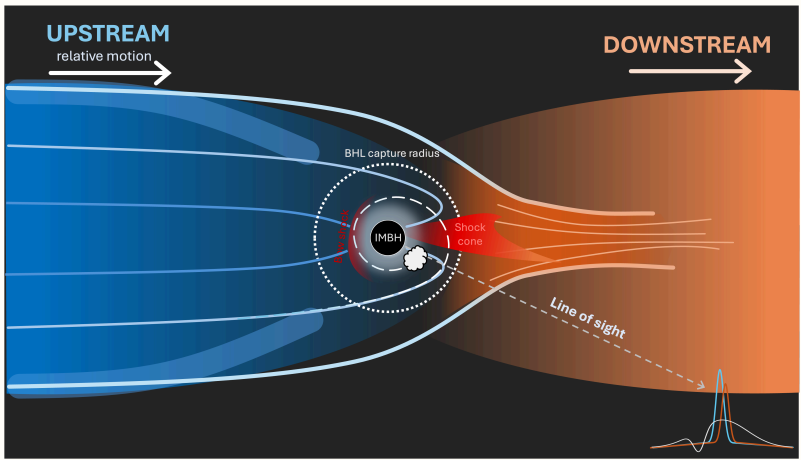


Figure 4 | Bondi-Hoyle-Lyttleton schematic shows gravitational focusing compressing upstream gas into a dense downstream accretion wake.